

Examiner reports

Q1.

Part (a) was done well by the majority of students. Most knew how to set up the initial differential equation and those who did generally scored full marks. There were a few numerical slips seen and some students did not complete the argument to show the required result.

A high proportion of students made good progress on part (b) with over half of students scoring 2 or 3 marks. Having correctly integrated, many students gave no consideration to the constant of integration and, although it was zero, its value needed to be found to correctly show the given result.

Part (c)(i) was well attempted and saw the full range of marks awarded. Over half of the students scored at least 4 marks with 6/6 being the most frequent score awarded. Many students realised that simultaneous equations had to be solved and usually eliminated a variable to form a correct equation. Students often made algebraic slips when solving their equation and a more efficient use of allowed calculator functions would have helped here.

In part (c)(ii) many students made some good criticisms of the equations they had found, but did not link these to the model used by the trader, which was the rate of sales is

proportional to $\frac{8-t}{x}$.

Q2.

The majority of students gained full marks on this graph sketch. A number of students lost marks for not labelling one or other of the intercepts, whilst quite a few had the vertex drawn at $(0, a)$ on the y -axis and therefore had no x -intercept. A small number of students did not know the basic shape of a linear modulus graph and therefore scored no marks.

Q4.

- (a) (i) Although there were still some who thought that the Factor Theorem involved long division, the vast majority found $p(3)$ and showed sufficient working to verify that $p(3) = 0$. There was still a significant number who failed to include a concluding statement such as “therefore $x - 3$ is a factor” and these students did not score full marks.
- (ii) The majority used long division or inspection to find the quadratic factor and then successfully factorised to be able to express $p(x)$ as a product of linear factors. The repeated factor seemed to disturb many students who felt that the factorised form must be something like $(x - 3)(x - 2)(x + 3)$.
- (b) A generous method mark was awarded for drawing a recognizable cubic curve. Those with the wrong factors usually earned this mark. Those with the correct factors were sometimes unsure of the correct shape of the curve in the vicinity of $x = 3$. It was pleasing to see many good sketches, but these were occasionally spoiled by not marking the values -2 and 3 on the x -axis or by poor curvature beyond $x = 3$ or before $x = -2$.

Q5.

In part (a), almost all the students gained the method mark for a relevant graph completely above the x -axis. To earn the accuracy mark, the students needed to indicate 4 on the x -axis, have a cusp there and correct curvature elsewhere. A small number of students did not gain the accuracy mark.

In part (b), the order of operations was crucial here. The great majority of the students recognised that a stretch and a translation were involved and most chose to describe

them in that order. However, these students rarely gave the correct vector $\begin{bmatrix} 0.5 \\ 0 \end{bmatrix}$ for the translation and lost one mark as a result. The small proportion of the students who gave the translation first were more likely to gain full marks.

Q6.

- (a) The students who understood the concept gained three marks easily, either by using two equations or by squaring both sides. A few solved all four relevant equations obtaining repeated values of x . However, it was disappointing to see many wrong approaches, particularly those which gave intercepts with the axes or answers such as $|x| = |3|$. The sketch was there to help; so values such as -3 should have been spotted as being clearly incorrect.
- (b) Many students spoiled their answers by giving combined inequalities such as $1^3 x^3 \leq 3$. Other incorrect responses included the interval $1 \leq x \leq 3$ or the use of " $x \leq 1$ and $x \geq 3$ ". Again, the sketch was there to assist, and it was good to see some students marking the diagram to help them choose the correct part.

Q7.

The responses to this question suggested that transformations and curve sketching need more attention.

- (a) A common incorrect response was to reflect the curve in the y -axis. Where students correctly reflected in the x -axis, errors occurred in indicating the intersection with the x -axis where answers of 6 and 18 were common. On the y -axis, $-\frac{4}{3}$ and -12 were often seen.
- (b) The most common sketch seen was a copy of the original curve with the part of the curve for $x < 0$ reflected back into the first quadrant. Of those who attempted to reflect in the y -axis, some failed to give a recognisable cusp at $(0, 4)$ and many reflected the 'ends' beyond ± 6 in the x -axis.
- (c) This part was better answered than the two previous parts. The main errors here were giving a stretch factor of -2 or $\frac{1}{2}$, or giving reflection in the x -axis or in the line $y = x$ instead of the y -axis. Students should note that "a horizontal stretch" is not sufficient and the answer should be given as "stretch parallel to the x -axis".

Q8.

In part (a)(i), those students who used long division, perhaps misunderstanding what was meant by 'The Factor Theorem', scored no marks. Most realised the need to find the value of $p(x)$ when $x = -1$. However, it was also necessary, after showing that $p(-1) = 0$, to write a concluding statement regarding $x + 1$ being a factor; some failed to show explicitly that $-(-5) = +5$ in their verification and lost a mark; a few evaluated $p(1)$ and scored no marks.

In part (a)(ii), those who found the quadratic factor using inspection were the most successful; methods involving long division or equating coefficients usually contained algebraic errors. Once a candidate had the correct quadratic factor, it was good to see most expressing $p(x)$ as the product of three correct factors. Others found the remaining factors separately by repeated use of the Factor Theorem, but then sometimes forgot to write $p(x)$ as the product of factors. A few tried to write down the product of factors with no working at all and, although some were successful, most using this direct approach made at least one error in their factors and scored no marks at all.

In part (b), it was not acceptable to write $p(0) > p(1)$ first, simply followed by $-6 > -8$; the verification required the correct evaluation of $p(0)$ and $p(1)$ and a concluding statement such as 'therefore $p(0) > p(1)$ '. Some made errors in their calculations, usually when finding $p(1)$, but the major reason for losing marks was the omission of a suitable conclusion.

In part (c), it was pleasing to see many correct graphs, although some failed to see the connection with part (b) and showed their minimum point either to the left of or actually on the y -axis. Some caused difficulties for themselves by marking the values of the intercepts $x = -1$ and $x = 2$ equidistant from the origin; the minimum point was then almost invariably drawn in the wrong place.

Q9.

There were many high marks on this question, although only a minority of students were able to answer part (b) correct.

Students answered part (a) well, with the majority giving a correct modulus graph. The main error was incorrect curvature in the outside branches $x > 3$ and $x < -2$.

In part (b), some students gave an identical response to the modulus graph in part (a), but the most common result was a correct plot for $x \geq 0$ but then the LHS of the curve was reflected in the y -axis to produce a sketch in the first and fourth quadrants.

There were many fully correct solutions to parts (c) and (d). The main errors were the use of the word transformation rather than translation and also the scale factor for the stretch stated as 2 in the x -axis rather than $\frac{1}{2}$ in the y -axis. The common errors in part (d) were (0,5) and (-1,20)

Q10.

This question was generally answered very well and full marks were often seen.

In part (a) many correct answers with correct notation were seen but there were many cases of $3 \leq f(x) \leq -3$ also seen. Where candidates lost a mark it was usually for poor notation.

In part (b)(i) most candidates earned the method mark for swapping x and y but the $\frac{1}{2}$ confused

many when trying to obtain $\cos^{-1} \frac{x}{3}$, with $\cos^{-1} \frac{2x}{3}$ and $\cos^{-1} \frac{x}{6}$ being common errors.

In (b)(ii) having the correct inverse function generally led to the correct answer here; unfortunately repeating incorrect algebra for part (b)(i) sometimes gave the 'right' answer, but obviously without reward.

Part (c)(i) was very well answered.

In part (c)(ii) most candidates achieved the method mark by drawing at least two continuous parts. Unfortunately many lost the final mark, as drawing multiple curves was a common error, as was labelling the x -axis incorrectly.

Part (d) was well done; a few candidates had $\frac{1}{3}$ or $\frac{1}{2}$ as the scale factor and a handful introduced another wrong transformation.

Q11.

A small number of candidates clearly did not recognise the modulus function at all.

In part (a)(i) most graphs were in the correct place and looked linear. However a few failed to extend into the first quadrant and many were lacking the values of either one or both intercepts and a few had the vertex labelled $-\frac{1}{3}$ or -3 .

In (a)(ii) although almost all graphs had the three requisite sections, quite a number had the outer parts linear or curving convexly instead of concavely. Many did not have all three intercepts correctly labelled.

In part (b)(i) the more able candidates were able to form two different quadratic equations and solve them, but many got only one pair of roots. However many formed wrong equations such as $3x + 3 = x^2 + 1$ as well and got extra roots. Having found correct roots some assumed that their negatives would also be roots, misunderstanding the modulus. Those who tried to square both sides usually made algebraic errors, and only a handful were able to establish correct roots from their quartic equation. Another misunderstanding about the modulus function caused some to discard the negative values that they had found.

In part (b)(ii) it was very common for the answers to be between their values rather than outside them. Only more able students produced the correct intervals.

Q12.

Part (a) was well answered by the majority of candidates. The main errors were not crossing the y -axis into the second quadrant or failing to give the points of contact with the axes.

In part (b), most candidates obtained the required two values. In most cases where candidates lost marks, it was for only giving one value, usually 2. Four values of $+2$, -2 ,

+6, and -6 was not uncommon.

Part (c) was not as well answered with many candidates trying to write down a single inequality for an answer.

Q13.

Part (a)(i) Those who used long division instead of the Factor Theorem scored no marks. Most candidates realised the need to show that $p(-3) = 0$. However quite a few omitted sufficient working such as $p(-3) = -27 + 63 - 21 - 15 = 0$, together with a concluding statement about $x + 3$ being a factor, and therefore failed to score full marks.

Part (a)(ii) Many candidates have become quite skilled at writing down the correct product of a linear and quadratic factor and then writing $p(x)$ as the product of three linear factors and these scored full marks. Others used long division or the Factor Theorem effectively but lost a mark for failing to write $p(x)$ as a product of linear factors. Others tried methods involving comparing coefficients, but often after several lines of working were unable to find the correct values of the coefficients because of poor algebraic manipulation. Speculative attempts to write down $p(x)$ immediately as the product of three factors were rarely successful.

Part (b)(i) This part of the question was intended to help candidates when sketching the curve. Those who found the correct values of $p(-1)$ and $p(0)$ usually scored the mark but in future a carefully written proof may be called for. Here again many arithmetic errors were seen.

Part (b)(ii) The sketch was intended to bring various parts of the question together but even very good candidates ignored the hint from part (b)(i) and showed a minimum point on the y -axis. A few lost the final mark when their curve stopped on the x -axis. Confusion between roots and factors spoiled many sketches and several showed the y -intercept of -15 on the positive y -axis. It was disappointing that many candidates did not recognize the shape of a cubic curve at all.

Q14.

(a) Attempts at this question varied from no attempt, through superficial attempts to some fully correct solutions. Most could separate the variables correctly, although a few did attempt "integrals" of $f(x)dt$. Some candidates inverted the given differential equation and so could attempt direct integration, which was a quite acceptable

method. Most integrated kdt correctly, but the integral of $(x+1)^{-\frac{1}{2}}$ defeated many. Often there was an error in the power, before or after integration, and also in the coefficient. However, most candidates did include a constant and attempted to find it from the given initial conditions, for which they gained credit. However, some did some algebraic manipulation before adding a constant, and so could gain no credit, as this was then incorrect. Many candidates who had correctly integrated the expressions failed to gain the last mark, which was for giving the answer in the form $x = f(t)$, because they couldn't square their expression correctly, often omitting the "middle" term.

(b) Candidates who had interpreted the context correctly could gain at least one mark here by substituting $t = 60$ into their result from part (a) and solving for x . However,

many confused the given $t = 60$ with the given initial condition $x = 80$. Relatively few correct answers were seen, with many candidates obtaining results of over 100% or indeed negative results. These were not apparently queried, with little evidence of checking for errors.

- (c) (i) Very few candidates wrote down a correct differential equation. Some who had seemingly understood the context, wrote down the correct expression in A , but did not include the constant of proportionality. Many candidates mixed up the variables with x , and less so y , appearing, although most had a “differential equation” of some sort. Some candidates even included units in their differential equation, which was condoned if at the end of a correct expression.
- (ii) For many candidates this was the only part of this question, they attempted. Candidates who had understood the context and started with $A = 4.5$ could mostly solve the equation for t successfully. The relatively rare error was in the algebraic manipulation or in attempts to take logarithms, such as $\log(1 + 4e^{-0.09t}) = \log 1 - 0.09 \log 4t$ or similar. Some made an error in miscopying -0.09 .

Those candidates who commonly started with $A = 50$ or $A = 0.5$ encountered the need to take the logarithm of a negative number in their working or obtained a negative value for t , which most then chose to ignore. They gained no credit. Some did realise their error in interpreting the context, and reverted to the correct value for A .

Q15.

Those candidates who used long division in part (a)(i) scored no marks. Most candidates realised the need to show that $p(-2) = 0$, but quite a few omitted sufficient working such as $p(-2) = -8 + 2 + 6 = 0$ together with a concluding statement about $x + 2$ being a factor and therefore failed to score full marks.

Many candidates have become quite skilled at writing down the correct product of a linear and quadratic factor and these scored full marks in part (a)(ii). Others used long division effectively but lost a mark for failing to write $p(x)$ in the required form. Others tried methods involving comparing coefficients, but often after several lines of working were unable to find the correct values of b and c because of poor algebraic manipulation.

In part (a)(iii), although many candidates tried to consider the value of the discriminant of their quadratic factor, quite a few used $a = 1$, $b = -1$ and $c = 6$ (from the cubic equation) and scored no marks for this part of the question. Others drew a correct conclusion using the quadratic equation formula, indicating that it was not possible to find the square root of -8 and others, after completing the square showed that the equation $(x - 1)^2 = -2$ has no real solutions. Some wrongly concluded that because it was not possible to factorise their quadratic then the corresponding quadratic equation had no real roots.

Most obtained the correct y -coordinate of B in part (b)(i).

In part (b)(ii) it was pleasing to see most candidates being able to integrate correctly but a large number did not answer the question set and simply found the indefinite integral in this part. Many candidates use poor techniques when finding a definite integral and it was often difficult to see the evaluation of $F(0) - F(-2)$ in their solution. Many obtained an answer of -10 which was miraculously converted into $+10$ with some comment about an

area being positive. This and similar dubious working was penalized.

In part (b)(iii) some obtained an answer of -6 for the area of the triangle by using -2 as the base. Credit was given to candidates who later realised that the area of the triangle was actually 6 . Unless candidates had scored full marks in part(ii) they were not able to score full marks in this part either, even if they obtained a correct value of 4 for the shaded area.

Q16.

In part (a)(i), a few candidates ignored the request to use the factor theorem and scored no marks for using long division. It was necessary to make a statement that " $x + 1$ is a factor", after showing that $f(-1) = 0$, in order to score full marks.

Part (a)(ii) was not answered as well as similar questions in previous years. Perhaps the sketch lured some into trying to write down three factors without any further working, rather than using the intermediate step of showing that $p(x) = (x + 1)(x^2 - x - 6)$ before writing $p(x)$ as a product of three factors. Many who tried long division were flummoxed by there being no x^2 term.

In part (b)(i), those who had the correct linear factors in part (a)(ii) usually wrote down correctly that A had coordinates $(-2, 0)$, although some carelessly wrote the point as $(0, -2)$.

Many candidates simply found an indefinite integral in part (b)(ii) and then a definite integral in part (b)(iii). The two parts were generously treated holistically when candidates did this. The fractions once again caused problems to most candidates who are so used to having a calculator to do this work for them. It was very rare to see the correct answer of -32 for the definite integral.

In part (b)(iii), many lost out on an easy mark because they rolled their two sections into one: those who wrote "integral $= -32 = 32$ " gained full credit for part (b)(ii) but did not score the mark in part (b)(iii). It was necessary to give a positive value for the area of the region and to make this explicit. In anticipation of a lot of wrong answers in part (b)(ii), a follow through mark was awarded in part (b)(iii): for example, if a candidate's answer in part (b)(ii) was -20 and they concluded that the area was 20 in part (b)(iii), they scored the mark.

In part (b)(iv), most candidates differentiated correctly, but quite a few thought that $3(-1)^2 - 7$ was equal to -10 and thus obtained the wrong gradient of the curve.

In part (b)(v), a large number of candidates found the correct equation of the normal but some still confused tangents and normals and consequently thought that the gradient of the normal was equal to -4 . It was quite common for weaker candidates to either negate their gradient or take the reciprocal but to fail to do both.

Q17.

Many candidates lost a mark in part (a)(i) for failing to make a statement regarding the factor: they assumed that it was sufficient just to show that $p(-2) = 0$.

In part (a)(ii), those who found the quadratic factor $x^2 - x - 6$ by long division usually went on to earn full marks. Some used the Factor Theorem but the repeated factor was problematic. Those who used inspection or comparing coefficients generally scored at